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Laguna State Polytechnic University
Province of Laguna

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Enterprise Infrastructure Planning for a Startup Company

Week 02 – BSIT 4A: System Administration and Maintenance

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Part 1 - Company Profile

Company Name: NovaStack Solutions

Nature of Business: NovaStack Solutions is a software development startup that builds custom web and mobile applications for small and medium-sized businesses. The company also offers basic IT consulting services to its clients, helping them modernize outdated systems.

Company Vision: To become a trusted technology partner for growing businesses by delivering reliable, scalable, and user-friendly software solutions.

Company Mission: NovaStack Solutions is committed to helping businesses solve real problems through well-designed software, while fostering a workplace built on innovation, collaboration, and continuous learning for its employees.

Office Location (fictional): 3rd Floor, Meridian Business Center, 88 Rizal Avenue, Calamba City, Laguna, Philippines

Organizational Structure:

- CEO / Managing Director
 - IT Department (5 employees) — handles infrastructure, technical support, and internal systems
 - Human Resources (4 employees) — handles recruitment, employee relations, and admin functions
 - Finance (5 employees) — handles budgeting, payroll, and financial reporting
 - Sales (6 employees) — handles client acquisition and account management

Employee Distribution:

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
Total	20



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Part 2 - Enterprise Hardware Inventory

Since NovaStack Solutions is starting completely from scratch, this inventory covers the essential hardware needed to get all four departments fully operational.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	10	IT (5), Finance (5)	Primary workstations for developers and finance staff who need higher processing power for coding, testing, and financial software
HW-002	Laptop	10	HR (4), Sales (6)	Portable devices for HR staff doing interviews/off-site work and Sales staff meeting clients outside the office
HW-003	Server	1	IT / Server Room	Hosts internal applications, file sharing, and acts as the domain/authentication server for the company
HW-004	Router	1	Network Room	Manages internet traffic and connects the internal network to the ISP
HW-005	Switch	2	Network Room / Office Floor	Distributes wired network connections to all departments
HW-006	Printer	4	One per department	Allows each department to print documents, contracts, and reports without sharing a single device across the floor
HW-007	UPS	3	Server Room, IT, Finance	Protects critical equipment (server, key workstations) from power outages and prevents data loss
HW-008	Wireless Access Point	2	Office Floor	Provides Wi-Fi coverage across the single office floor for laptops and mobile devices
HW-009	NAS Storage	1	IT / Server Room	Centralized storage for shared files, backups, and department documents
HW-010	External Backup Drive	2	IT	Used for offline backups of critical company and server data as a secondary backup layer
HW-011	Monitor	10	IT (5), Finance (5)	Paired with desktop computers for dual-purpose or extended display needs

Justification Summary: Desktops were assigned to IT and Finance since these roles benefit from stationary, higher-performance setups (development environments, spreadsheets, financial systems), while laptops went to HR and Sales for mobility during interviews and client visits. One server and one NAS unit are sufficient at this company size but leave room to scale. UPS units are prioritized for the server and finance systems since those are the most damage-prone if power is interrupted mid-task.



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Part 3 - Enterprise Software Inventory

This section covers the core software NovaStack Solutions needs across development, productivity, security, and system administration.

Software	Version	License	Purpose
Windows 11 Pro	23H2	Volume License (per device)	Main OS for staff workstations and laptops; Pro edition needed for domain join, BitLocker, and remote management features
Ubuntu Server	22.04 LTS	Free / Open Source	Runs on the company server for hosting internal apps, file sharing, and development/testing environments
Microsoft Office	Microsoft 365 Business Standard	Subscription (per user)	Word, Excel, Outlook, and Teams for daily documentation, reporting, communication, and collaboration
VS Code	1.9x (latest stable)	Free / Open Source	Primary code editor for the IT/development team to write and debug software
Git	2.4x	Free / Open Source	Version control system used by developers to track code changes and collaborate on projects
GitHub Desktop	Latest stable	Free	Simplifies Git operations with a visual interface, useful for less command-line-comfortable staff
VirtualBox	7.x	Free / Open Source	Allows IT staff to run virtual machines for testing software, OS configurations, and training environments safely
Google Chrome	Latest stable	Free	Standard web browser for daily internet access and web-based tools across all departments
Microsoft Defender	Included with Windows 11 Pro	Free (bundled)	Built-in antivirus and endpoint protection to guard against malware and basic threats
AnyDesk	Latest stable	Free (Business license for IT)	Remote desktop tool for IT support to troubleshoot employee devices without being physically present
7-Zip	23.x	Free / Open Source	File compression and extraction tool used company-wide for handling large files and archives

Why this matters: Every software choice here balances cost and functionality — many tools (Ubuntu Server, VS Code, Git, VirtualBox, 7-Zip) are free and open-source, which keeps licensing costs low for a startup, while paid tools like Windows 11 Pro and Microsoft 365 are used where business-grade features (security, collaboration, support) are actually necessary. Microsoft Defender is included at no extra cost and is sufficient as a baseline antivirus for a company this size, avoiding an unnecessary third-party licensing expense in the early stages.



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Part 4 - Enterprise Network Inventory

This covers the networking equipment needed to physically connect and secure NovaStack Solutions' office network.

Asset ID	Equipment	Quantity	Purpose
NW-001	ISP Modem	1	Connects the office to the internet service provider's network
NW-002	Router	1	Directs traffic between the internet and the internal office network; assigns internal IP addresses
NW-003	Firewall	1	Filters incoming and outgoing traffic to protect the network from unauthorized access and cyber threats
NW-004	Switch	2	Connects wired devices (desktops, printers, server) within each department to the network
NW-005	Wireless Access Point	2	Extends Wi-Fi coverage across the office floor for laptops and mobile devices
NW-006	Patch Panel	1	Organizes and terminates all CAT6 cable runs from workstations into a central, manageable point
NW-007	CAT6 Cables	50 meters (bulk roll)	Provides wired connections between workstations, switches, and the patch panel with gigabit speed support
NW-008	RJ45 Connectors	100 pieces	Used to terminate CAT6 cables into functional network connectors for patch panel and device ports

Notes: The firewall is placed right after the router as a dedicated security layer rather than relying only on the router's built-in firewall — this gives NovaStack Solutions better control over traffic rules as the company grows. Two switches are enough to cover all four departments on a single floor, and the patch panel keeps cable management clean and easy to troubleshoot instead of running cables directly from every desk to the switch.

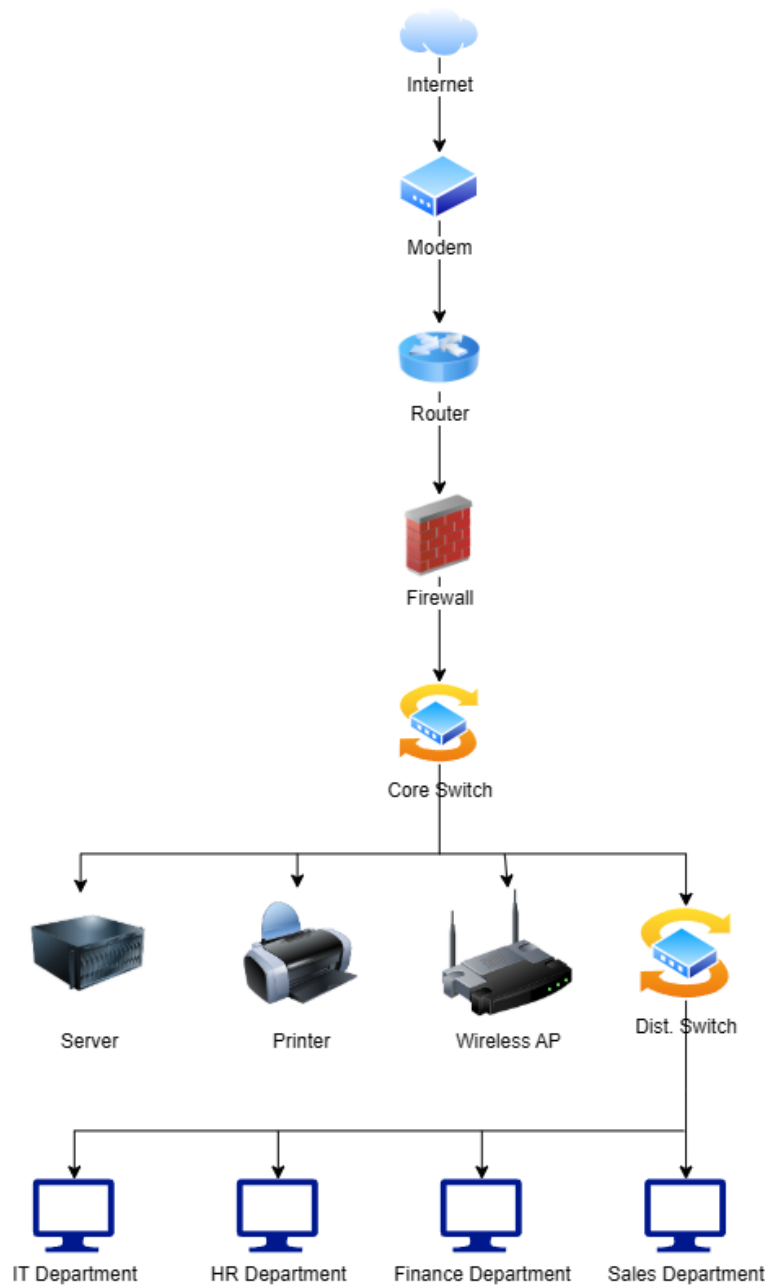


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Part 5 - Enterprise Network Diagram

NovaStack Solutions Enterprise Network Topology



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Part 6 - System Administration Roles

Helpdesk Technician

Responsibilities: Handles day-to-day technical support requests from employees, resolves hardware and software issues, sets up new workstations, resets passwords, and escalates more complex problems to senior IT staff.

Skills: Strong troubleshooting ability, patience, clear communication with non-technical users, basic knowledge of Windows/Mac OS, and familiarity with common office software.

Common Tools: Remote desktop software (like AnyDesk or TeamViewer), ticketing systems (e.g., Zendesk, Freshdesk), Microsoft Office, basic diagnostic tools.

Certifications: CompTIA A+, Microsoft Certified: Modern Desktop Administrator Associate.

Network Administrator

Responsibilities: Manages and maintains the company's network infrastructure — routers, switches, firewalls, and wireless access points. Monitors network performance, configures IP addressing, and troubleshoots connectivity issues.

Skills: Deep understanding of networking protocols (TCP/IP, DNS, DHCP), subnetting, VLANs, and network security fundamentals.

Common Tools: Cisco Packet Tracer, Wireshark, SolarWinds, PRTG Network Monitor.

Certifications: CompTIA Network+, Cisco CCNA.

Linux System Administrator

Responsibilities: Installs, configures, and maintains Linux-based servers (such as the Ubuntu Server used by NovaStack Solutions). Manages user permissions, automates tasks with scripts, and ensures server uptime and security.



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Skills: Command-line proficiency, shell scripting (Bash), knowledge of file systems and permissions, package management, and basic security hardening.

Common Tools: Terminal/SSH, cron, Bash scripting, Ansible, Docker.

Certifications: CompTIA Linux+, Red Hat Certified System Administrator (RHCSA).

Cloud Administrator

Responsibilities: Manages the company's cloud-based resources — including backups, virtual servers, and storage — and ensures cloud services scale properly as the company grows. Monitors cloud costs and security configurations.

Skills: Familiarity with cloud platforms (AWS, Azure, or Google Cloud), understanding of virtualization, cost management, and cloud security best practices .

Common Tools: AWS Management Console, Microsoft Azure Portal, Google Cloud Console, Terraform.

Certifications: AWS Certified Cloud Practitioner, Microsoft Certified: Azure Administrator Associate.

How these four roles work together: In an organization like NovaStack Solutions these four professionals do not work alone. They form a support chain that keeps the infrastructure running smoothly. The Helpdesk Technician is usually the point of contact for employees handling everyday issues and escalating anything beyond basic troubleshooting. When a problem involves connectivity or hardware, at the network level it gets passed to the Network Administrator, who manages the routers, switches and firewalls that keep the office connected. Meanwhile the Linux System Administrator keeps the backend server running ensuring internal applications and files remain accessible and works closely with the Cloud Administrator especially as the company starts relying on cloud backups or remote services for scalability. Together they form support: frontline help, network stability, server management and cloud scalability all working in coordination to keep the business operational.



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Part 7 - Infrastructure Recommendations

Internet Provider: NovaStack Solutions should subscribe to a business-grade fiber internet plan (e.g., PLDT Business Fiber or Converge Business) with at least 100 Mbps symmetrical speed. Business plans typically include better uptime guarantees (SLA), static IP options, and faster support response times compared to residential plans — both of which matter for a company running its own server and relying on stable connectivity for daily operations.

Server Specifications: The company server should run at minimum an Intel Xeon or AMD EPYC processor, 32GB RAM, and RAID-configured storage (RAID 1 or RAID 5) for redundancy. Since the server hosts internal applications and shared files for 20 employees, it needs enough processing power and memory to handle multiple simultaneous connections without slowing down, while RAID protects against data loss if a single drive fails.

Backup Strategy: A 3-2-1 backup approach is recommended — 3 copies of data, stored on 2 different media types, with 1 copy off-site (e.g., cloud storage). In practice, this means the NAS storage handles daily local backups, the external backup drive provides a secondary on-site copy, and a cloud backup service covers the off-site requirement in case of fire, theft, or hardware failure at the office.

Security Recommendations: Beyond the firewall already in the network design, NovaStack Solutions should implement network segmentation (VLANs) to separate department traffic, enable WPA3 encryption on the wireless network, and require multi-factor authentication (MFA) for access to sensitive systems like the server and financial software. These layers reduce the risk of a single compromised device affecting the entire network.

Antivirus: Microsoft Defender (already included with Windows 11 Pro) is sufficient as a baseline for this company size, since it offers real-time protection without additional licensing costs. As the company grows, upgrading to a centrally managed endpoint solution (e.g., Microsoft Defender for Business) would allow IT to monitor threats across all devices from one dashboard.

Password Policy: Enforce a minimum 12-character password length with a mix of uppercase, lowercase, numbers, and symbols, along with mandatory password changes every 90 days and account lockout after 5 failed login attempts. This reduces the risk of brute-force attacks and unauthorized access, especially for accounts with access to the server or financial systems.



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Expansion Plan: As NovaStack Solutions grows beyond 20 employees, the network should be designed to scale by adding additional switches and access points rather than replacing core infrastructure. The server should also support virtualization (via VirtualBox or a hypervisor) so additional services can be added without needing new physical hardware right away, keeping expansion costs manageable in the early growth stages.

Part 8: Personal Reflection

Going into this project, I honestly thought planning an IT infrastructure would just mean listing a bunch of hardware and calling it done. But working through all eight parts of this plan for NovaStack Solutions made me realize how much thinking actually goes into a network before a single device gets plugged in. One of the biggest things I learned was how interconnected every decision is. Deciding how many employees were in each department wasn't just for the company profile — it directly affected how many desktops and laptops I needed, which affected how many switches I needed, which then affected how the network diagram had to branch out logically. I couldn't just fill things in randomly; every number and choice had to make sense with the ones before it.

The task I found most challenging was the network diagram. At first, I thought I could just connect every device straight down to a department, but that didn't reflect how real networks are structured. I had to rethink it a few times — realizing that shared devices like the server, printer, and wireless AP shouldn't be tied to a single department, and that a second switch was needed to properly distribute the connection to IT, HR, Finance, and Sales. It took a few tries to get the logic right, but it helped me actually understand why real office networks are built with a core switch and distribution switch instead of one switch handling everything.

Planning before deployment matters because fixing a mistake on paper costs nothing, but fixing a mistake after equipment is already bought and wired is expensive and time-consuming. If I had skipped straight to buying hardware without mapping out the departments, network flow, and software needs first, NovaStack Solutions could have ended up with the wrong quantities, security gaps, or a network that doesn't scale well as the company grows.

This project helped me see system administration less as "fixing things when they break" and more as a role that requires foresight — anticipating what a company will need, not just reacting to problems. Working through the



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hardware, software, and network inventories together also showed me how a Junior System Administrator has to think across departments, not just from an IT-only perspective. I feel more confident now in how I'd actually approach building out infrastructure for a real company, starting from nothing.